

Please check the information details below before entering your candidate information

Centre Number	Candidate Name
	Nikhil

Centre Number	Candidate Number

Pearson Edexcel International GCSE

Thursday 16 May 2024

Morning Time 2 hours	Paper reference
	4MA1/1H

Mathematics A PAPER 1H Higher Tier	
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You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser, calculator. <i>Working paper may be used.</i>	Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided.
– There may be more space than you need.
- Calculators may be used.
- You must **NOT** write anything on the formulae page.
Anything you write on the formulae page will gain **NO** credit.

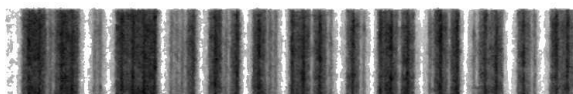
Information

- The total mark for this paper is 100.
- The marks for each question are shown in brackets.
– Use this as a guide as to how much time to spend on each question.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ►



Answer ALL TWENTY FIVE questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 Here are the first four terms of an arithmetic sequence.

1 4 7 10 n

- (a) Find an expression, in terms of n , for the n th term of this sequence.

$$T_n = a + (n-1)d$$

~~$$T_n = 1 +$$~~

$$T_n = 1 + (n-1)3$$

$$= 1 + 3n - 3$$

$$= 3n - 2$$

$$T_5 = 3 \times 5 - 2$$

$$= \underline{\underline{13}}$$

17
(2)

The n th term of a different arithmetic sequence is $5n + 17$

- (b) Find the 12th term of this sequence.

$$T_{12} = 5 \times 12 + 17$$

$$= 60 + 17$$

$$= \underline{\underline{77}}$$

77

(1)

(Total for Question 1 is 3 marks)

- 2 450 students were asked how they travelled to school on Monday.
 Each student walked or travelled by bus or travelled by car or travelled by bicycle.
 Each student used just one method of travel.

One of these students is chosen at random.

The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

$$0.20 + x + 2x + 0.26 = 1$$

$$3x + 0.46 = 1$$

$$3x = 1 - 0.46$$

$$\frac{3x}{3} = \frac{0.54}{3}$$

$$x = 0.18$$

$$\begin{aligned} \therefore \text{Car} &= 2x \\ &= 2 \times 0.18 = 0.36 \end{aligned}$$

$$\begin{aligned} \therefore \text{No of students travelling} &= 0.36 \times 450 \\ \text{by car} &= \underline{\underline{162}} \end{aligned} \quad 162 \text{ students}$$

(Total for Question 2 is 4 marks)

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

- 3 Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

$$\begin{array}{r} 2 \overline{) 72} \\ 2 \overline{) 36} \\ 2 \overline{) 18} \\ 3 \overline{) 9} \\ 3 \overline{) 3} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 108} \\ 2 \overline{) 54} \\ 3 \overline{) 27} \\ 3 \overline{) 9} \\ 3 \overline{) 3} \\ 1 \end{array}$$

$$72 = 2 \times 2 \times 2 \times 3 \times 3$$
$$108 = 2 \times 2 \times 3 \times 3 \times 3$$

$$\text{HCF} = 2^2 \times 3^2$$
$$= \underline{36}$$

36

(Total for Question 3 is 2 marks)

- 4 Ava records the number of kilometres she drives each month.

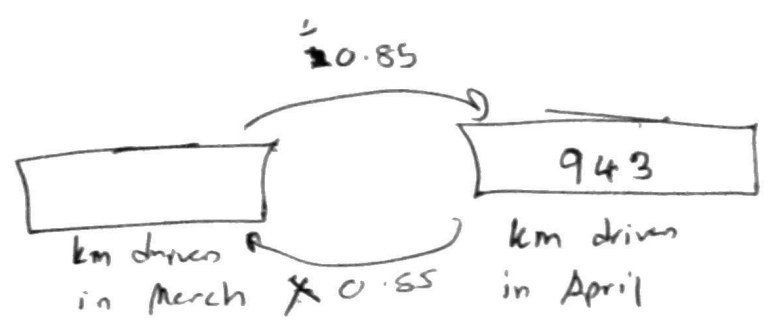
In April, Ava drove 943 kilometres.

This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

$$\frac{100 - 15}{100}$$

$$= 0.85$$



$$\left. \begin{array}{l} \text{No. of k'l'o m} \\ \text{driven in March} \end{array} \right\} = \frac{943}{0.85}$$

$$= 801.515$$

$$= \underline{801.6}$$

801.6 kilometres

(Total for Question 4 is 3 marks)

5 In the diagram, $ABCDE$ is a regular pentagon.

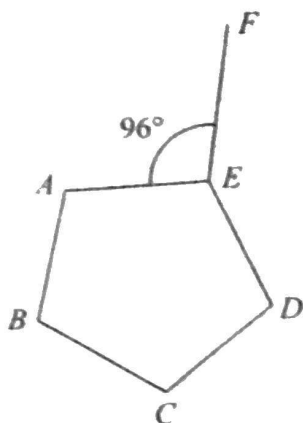


Diagram NOT accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

$$\begin{aligned} \text{Total Interior } \angle &= 180(n-2) \\ &= 180(5-2) \\ &= 540 \end{aligned}$$

$$\begin{aligned} \therefore \text{Individual interior } \angle &= \frac{540}{5} \\ &= 108 \end{aligned}$$

$$\therefore \angle AED = 108$$

$$96 + 108 + \angle FED = 360 \text{ (}\angle\text{s around a point)}$$

$$204 + \angle FED = 360$$

$$\begin{aligned} \angle FED &= 360 - \cancel{204} \\ &= \underline{156} \end{aligned}$$

156°

(Total for Question 5 is 4 marks)

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DO NOT WRITE IN THIS AREA

- 6 (a) Expand and simplify $(m+5)(m-8)$

$$m(m-8) + 5(m-8)$$

$$m^2 - 8m + 5m - 40$$

$$\underline{\underline{m^2 - 3m - 40}}$$

$$m^2 - 3m - 40$$

(2)

- (b) Solve $3n - 4 = \frac{5n + 6}{3}$

Show clear algebraic working.

$$3 \times 3n - 4 = \frac{5n + 6}{3} \times 3$$

$$5n + 6 = 3(3n - 4)$$

$$5n + 6 = 9n - 12$$

$$9n - 5n = 12 + 6$$

$$\frac{4n}{4} = \frac{18}{4}$$

$$\underline{\underline{n = 4.5}}$$

$$n = 4.5$$

(2)

(Total for Question 6 is 5 marks)

7 $E = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$

$A = \{\text{even numbers}\}$

$B = \{23, 29, 31\}$

$C = \{\text{multiples of 3}\}$

$A = \{24, 26, 28, 30, 32, 34\}$

$B \cap C = \{25, 30, 33\}$

(a) List the members of the set

(i) $B \cup C$

$\{23, 25, 29, 30, 31, 33\}$

(ii) $A' \cap C$

$\{25, 30, 33\}$

(b) Is it true that $B \cap C = \emptyset$?

Tick (✓) one of the boxes below.

Yes ☒

No ☐

Give a reason for your answer.

This is because, there is no multiple of 3 in B.

The set D has 4 members and is such that $D \cap (A \cup C) = \emptyset$

(c) List the members of set D

(Total for Question 7 is 5 marks)

- 8 A cylinder is placed on a table.

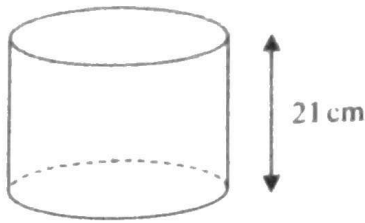


Diagram **NOT** accurately drawn

The volume of the cylinder is 1575 cm^3

The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

$$\text{Volume of Cylinder} = \pi r^2 h$$

$$\frac{1575}{\pi \cdot 21} = \frac{\pi r^2 \cdot 21}{\pi \cdot 21}$$

$$r^2 = \frac{1575}{\pi \cdot 21}$$

$$r = \sqrt{\frac{1575}{21\pi}}$$

$$r = 4.886$$

$$\begin{aligned} \therefore \text{Area of Cylinder} &= 2\pi r h + 2\pi r^2 \\ &= 2 \times \pi \times 4.886 \times 21 + 2 \times \pi \times 4.886^2 \\ &= 794.690 \end{aligned}$$

$$0.106 \text{ newtons/cm}^2$$

$$\therefore \text{Pressure} = \frac{84}{794.690}$$

$$= 0.1057$$

$$= \underline{0.106 \text{ newtons/cm}^2}$$

(Total for Question 8 is 3 marks)

- 9 The table gives the amount of rice produced by each of two countries in 2020

Country	Amount of rice (tonnes)
Indonesia	3.5×10^7
Argentina	8.2×10^5

- (a) Write 3.5×10^7 as an ordinary number.

35000000
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

Argentina $\rightarrow 820000$

820,000

$$\therefore \text{Japan} = 6,780,000 + \cancel{820,000}$$

$$= 7660000$$

$$= \underline{\underline{7.6 \times 10^6}}$$

7.6×10^6 tonnes
(2)

(Total for Question 9 is 3 marks)

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10 (a) Simplify $(2p)^9$ where $p > 0$

(1)

$$y^a \times y^b = y^c$$

(b) Find the value of n

$$\begin{aligned} n &= 9 + -3 \\ &= 9 - 3 = \underline{\underline{6}} \end{aligned}$$

$$n = 6$$

(1)

(c) Simplify fully $(5a^4c^2)^3$

$$\begin{aligned} &5^3 a^{4 \times 3} c^{2 \times 3} \\ &= \underline{\underline{125 a^{12} c^6}} \end{aligned}$$

$$125 a^{12} c^6$$

(2)

(Total for Question 10 is 4 marks)

11 The diagram shows a roof support.

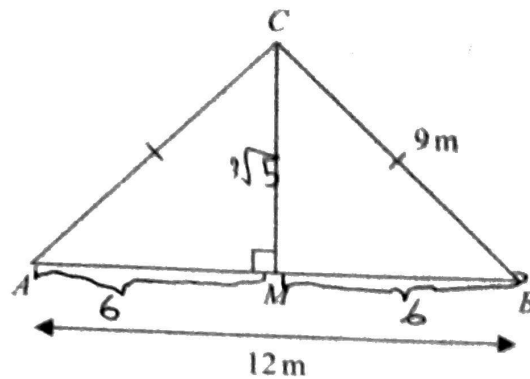


Diagram NOT
accurately drawn

The roof support is made from four lengths of wood, AB, AC, BC and MC

$$AC = BC = 9\text{ m} \quad AB = 12\text{ m}$$

$$\text{angle } AMC = 90^\circ$$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.

Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.
Show your working clearly.

M is in between AB

$$AM = MB = \frac{12}{2} = 6$$

Pythagoras Theorem,

$$CB^2 = CM^2 + MB^2$$

$$CM = \sqrt{9^2 - 6^2} = 3\sqrt{5}$$

$$\begin{aligned} \therefore \text{Perimeter of wood} &= 12 + 9 + 3\sqrt{5} + 9 \\ &= 30 + 3\sqrt{5} \text{ m} \end{aligned}$$

$$\therefore \text{total cost} = 21.50 \times 30 + 3\sqrt{5}$$

$$= 789.276$$

$$= \underline{\underline{€789.23}}$$

789.23 euros

(Total for Question 11 is 4 marks)



12 (a) Factorise fully $6y^2 - 5y - 4$

$$6y^2 + 3y - 8y - 4$$

$$3y(2y+1) - 4(2y+1)$$

$$(2y+1)(3y-4)$$

$$\underline{y = -\frac{1}{2} \text{ or } y = \frac{4}{3}}$$

$$6y^2 \times 4$$

$$= 24y^2$$

$$\begin{array}{l} / \quad \backslash \\ -8y \quad +3y \end{array}$$

$$y = -\frac{1}{2} \text{ or } y = \frac{4}{3}$$

(2)

(b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.

$$\frac{3x \times 2x + 1}{3x \times 4x} + \frac{7 - 5x \times 4x}{3x \times 4x}$$

$$\frac{3x(2x+1) + 4x(7-5x)}{12x^2}$$

$$\frac{6x^2 + 3x + 28x - 20x^2}{12x}$$

$$\frac{-20x^2 + 6x^2 + 3x + 28x}{12x}$$

$$\frac{-14x^2 + 31x}{6x}$$

$$\underline{\underline{-\frac{7x^2 + 31}{6}}}$$

$$-\frac{7x^2 + 31}{6}$$

(3)

(Total for Question 12 is 5 marks)